

Proposed Solution

Project: Rising Water – AI-Based Flood Prediction System

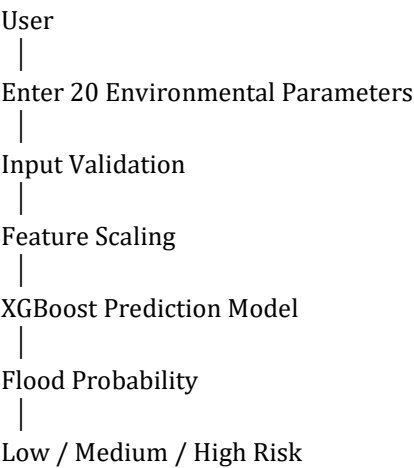
1. Solution Overview

Rising Water is a web-based Machine Learning application that predicts flood probability from 20 environmental parameters. It uses an XGBoost model trained on historical flood data and provides an instant flood risk assessment.

2. Objectives

- Predict flood probability accurately.
- Provide an easy-to-use web interface.
- Support early disaster preparedness.
- Reduce potential loss through timely prediction.

3. System Workflow



4. Technology Stack

Technology	Purpose
Python	Programming Language
Pandas	Data Processing
NumPy	Numerical Computing
Scikit-learn	Preprocessing
XGBoost	Prediction Model
Flask	Backend Framework
HTML/CSS	Frontend
Joblib	Model Storage

5. Expected Benefits

- Early flood warning
- Improved decision making

- Reduced property and crop loss
- Simple and accessible interface
- Future integration with weather APIs

6. Conclusion

The proposed solution combines Machine Learning and a Flask web application to deliver a practical, scalable, and user-friendly flood prediction system.